

NARDL Summary from RW4 Primary Chain

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

RW4 domestic vertical chain is FarmGateUA -> ProducerUA -> ProZorro -> Retail, estimated with both forward and reverse-flow pairs. Farm-gate enters from two alternative reconstruction workbooks and both linear/pchip interpolation variants are carried end-to-end. Retail_combined is the anchored downstream index built from Silpo effective prices, Novus observed prices, and a level-aligned ConsumerUA anchor, while Retail_combined_core keeps the strict retailer-only overlap for robustness.

family=NARDL

rows=3028

source=/Users/getapple/Documents/KSE/Master Thesis/Main materials/Model/Charniuk_Dairy_Research/outputs/primary_chain_summary/primary_chain_consolidated.xlsx

Table: NARDL_Summary

panel_level	panel_name	segment_name	product	standardized_type	retailer_panel	brand	price_variant	reconstruction_variant	farm_gate_source	frequency	intersection_rule	pair_scope	chain_direction	stage_from	stage_to	link_model_family	y_series	x_series	n_obs	sr_coef	lr_coef	ect_coef	ect_pvalue	asymmetry_short_p	asymmetry_long_p	vecm_rank	model_status	unreliable_flag	unreliable_reason	shock_dummy_included	promo_controls_included	ljungbox_p	arch_p	jb_p	notes	robust_linear_vs_pchip	interpolation_sensitive	robust_across_reconstruction	core_finding_flag			
pairwise_product	pairwise::farm_gate_produ...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producuerua	NARDL	ProducerUA	FarmGateUA	1458	0.7536110832625593	1.266815309357476	-0.01240042740251713	1.557032310570778e-11	7.308663215981137e-05	0.02532017267918595	nan	unreliable	1	residual_autocorr_or_arch	0	0	2.541208735492714e-56	1.029527901378011e-56	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producuerua_to_farmgateua	NARDL	FarmGateUA	ProducerUA	1458	0.05147636888710523	0.0855496292369927372	-0.0032311348666763982	0.009291912187040789	0.7994265967946689	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.603167599167287e-11	9.836640713246854e-47	0.0	Wald	short/long	asymmetry...	0	1	0	0	
pairwise_product	pairwise::farm_gate_produ...	Кефір	Кефір	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producuerua	NARDL	ProducerUA	FarmGateUA	1458	0.1564641180836793	0.6096904480686457	-0.001534557185330408	8.803341208758638e-14	0.007783580888595607	0.1267443035147452	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.921289963727872e-55	1.611013527456772e-49	0.0	Wald	short/long	asymmetry...	0	1	0	0
pairwise_product	pairwise::farm_gate_produ...	Кефір	Кефір	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	farmgateua_to_producuerua	NARDL	FarmGateUA	ProducerUA	1458	0.03707747269361752	-0.05361011087413249	-0.003228379414410878	0.0392799735677546	0.01837307788005506	0.8796832708381467	nan	unreliable	1	residual_autocorr_or_arch	0	0	2.165780843478571e-11	9.13928801251949e-48	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producuerua	NARDL	ProducerUA	FarmGateUA	1458	0.3842652369068096	0.007493940665739711	0.01283113500623241	1.161901661881545e-13	0.008628192353514896	0.986344853053989	nan	unreliable	1	residual_autocorr_or_arch	0	0	2.216113100659389e-55	2.238530037089499e-47	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producuerua_to_farmgateua	NARDL	FarmGateUA	ProducerUA	1458	0.03103902358068476	-0.0586985080570141	-0.00330489804604967	0.03742859000646596	0.03341273956130519	0.862993804995912	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.003997386066738e-11	3.849649280495084e-49	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producuerua	NARDL	ProducerUA	FarmGateUA	1458	0.6495515851199283	0.6682757002312711	-0.0145226212209655	1.088862073698824e-13	0.0003162794868862343	0.1426981970130435	nan	unreliable	1	residual_autocorr_or_arch	0	0	8.4114813100242e-57	4.501458520956465e-51	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producuerua_to_farmgateua	NARDL	FarmGateUA	ProducerUA	1458	0.0434943155653237	-0.08759476497936694	-0.00329553349977777	0.04511296397304265	0.01451316515550449	0.7962036987805751	nan	unreliable	1	residual_autocorr_or_arch	0	0	2.05070000378277e-11	1.11353006009904e-47	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	forward	FarmGateUA	ProducerUA	farmgateua_to_producuerua	NARDL	ProducerUA	FarmGateUA	1458	0.5917718398199494	0.6330693925277114	-0.01388342787302217	2.029784178423845e-12	0.001666508027668	0.1749211426935455	nan	unreliable	1	residual_autocorr_or_arch	0	0	1.517697524813312e-54	1.156974678575467e-46	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_produ...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_producer	reverse	ProducerUA	FarmGateUA	producuerua_to_farmgateua	NARDL	FarmGateUA	ProducerUA	1458	0.042944320837532414	-0.1437890807485015	-0.003274111720239434	0.042044183367780e-12	0.012619573208197094	0.6865565583759783	nan	unreliable	1	residual_autocorr_or_arch	0	0	4.054826626980248e-11	2.594045418336778e-47	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_prozo...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	NARDL	ProZorro	FarmGateUA	220	-8.331652463361861	-0.9154775416329228	-0.8024349207088957	3.476047899864406e-26	0.3576494934013459	0.7272154293568747	nan	ok	0	nan	0	0.1067209530412837	0.6216059780109093	4.166216144037865e-57	Wald	short/long	asymmetry...	1	1	0	0	
pairwise_product	pairwise::farm_gate_prozo...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	NARDL	FarmGateUA	ProZorro	220	-0.00104484965154605	-0.02468198864082874	-0.05215455439220817	0.0004074704633854786	0.535299508075095	0.57820747699896089	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.9713094719379907	0.0449477699896089	5.353885733154511e-33	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::farm_gate_prozo...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	NARDL	ProZorro	FarmGateUA	216	-5.615722131190427	-0.8583830089738245	-1.108868845250867	3.746246796705595e-42	0.6920478412371363	0.5197982861063085	nan	ok	0	nan	0	0.5475364696507483	0.6975834725210998	0.0	Wald	short/long	asymmetry...	1	1	0	0	
pairwise_product	pairwise::farm_gate_prozo...	Молоко питне	Молоко питне	milk	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	NARDL	FarmGateUA	ProZorro	216	-0.003320092527215861	-0.1360073408251827	-0.05774554446866729	0.0003860672879006522	0.1053639845420947	0.01135320213905949	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.7002144147281233	0.0001525872213691102	2.482331698580671e-82	Wald	short/long	asymmetry...	1	1	0	0
pairwise_product	pairwise::farm_gate_prozo...	Сир кисломолочний	Сир кисломолочний	cottage_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	NARDL	ProZorro	FarmGateUA	181	-7.24488988577756	1.129395405138077	-0.7781788713321292	2.097515180804024e-17	0.5011921959545087	0.7588650654827862	nan	ok	0	nan	0	0.8259607920032	0.103590181728141	0.6577519881122729	Wald	short/long	asymmetry...	1	1	0	0	
pairwise_product	pairwise::farm_gate_prozo...	Сир кисломолочний	Сир кисломолочний	cottage_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	NARDL	FarmGateUA	ProZorro	181	-0.00101723604488973	-0.01055865637330252	-0.07579060777351333	0.0006328474730782903	0.4637039487906589	0.6457806572921767	nan	ok	0	nan	0	0.1262989390601314	0.1586188110977716	2.40724035480198e-19	Wald	short/long	asymmetry...	1	1	0	0	
pairwise_product	pairwise::farm_gate_prozo...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	NARDL	ProZorro	FarmGateUA	189	-17.36997063027372	-0.5157506849232467	-0.6962948940425042	1.091845492756672e-15	0.005986514377680486	0.8833074673807098	nan	ok	0	nan	0	0.483526286322754	0.7287829726035183	0.000791688118138595	Wald	short/long	asymmetry...	1	1	0	0	
pairwise_product	pairwise::farm_gate_prozo...	Сир твердий	Сир твердий	hard_cheese	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	NARDL	FarmGateUA	ProZorro	189	-0.00421492637556673	-0.06469153540894483	-0.06502314765300099	0.001776573241197639	0.01870240978986244	0.03167379328868442	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.05476567020747605	0.0007689702513166254	3.963850806817059e-36	Wald	short/long	asymmetry...	1	1	0	0
pairwise_product	pairwise::farm_gate_prozo...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	NARDL	ProZorro	FarmGateUA	178	-0.0446785871386613	1.103127276910245	-0.906074480975449	4.53582398628733e-25	0.9237491725490369	0.65923389051882742	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.043088905367281742	0.04525689642625472	0.04525689642625472	Wald	short/long	asymmetry...	1	1	0	0
pairwise_product	pairwise::farm_gate_prozo...	Сметана	Сметана	sour_cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	farm_gate_prozorro	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	NARDL	FarmGateUA	ProZorro	178	-0.001078607860401676	-0.01025129376714782	-0.07854963606428816	0.0007900141712274095	0.5732182857519146	0.8371831480242596	nan	ok	0	nan	0	0.1384763917275051	0.33815140662248897	1.315885028695882e-79	Wald	short/long	asymmetry...	1	0	0	0	
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProducerUA	farmgateua_to_producuerua	NARDL	ProducerUA	FarmGateUA	218	-0.6757439311909024	-4.482933812253403	-0.01464556076787184	0.0137470945278689	0.6498975669619452	0.4648865836719452	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.04895340806426387	2.6224618063819e-06	0.0	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	ProducerUA	ProZorro	producuerua_to_prozorro	NARDL	ProZorro	ProducerUA	218	-0.1474333602421413	0.2581457420186793	-0.8115030581919576	6.701748050261256e-28	0.8623480006513561	0.2290746927525529	nan	ok	1	nan	0	0.2721476798419011	0.6238213504277711	3.368059742284993e-77	Wald	short/long	asymmetry...	1	1	0	0	
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	forward	FarmGateUA	ProZorro	farmgateua_to_prozorro	NARDL	ProZorro	FarmGateUA	218	-9.8485550579512721	-1.02139095490055	-0.802940036974870754e-26	3.509836974870754e-13	0.6971516701681586	0.586024838557136	nan	ok	1	nan	0	0.0938354629043362	0.586024838557136	1.60545703615458e-54	Wald	short/long	asymmetry...	1	1	0	0	
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	reverse	ProZorro	ProducerUA	prozorro_to_producuerua	NARDL	ProducerUA	ProZorro	218	0.001737762578598349	-0.663127252462147	-0.01375165422973445	0.01827750354709647	0.7971132286260633	0.431149411685102	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.06619917256038729	6.971074870809132e-06	0.0	Wald	short/long	asymmetry...	1	1	0	0
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	reverse	ProducerUA	FarmGateUA	producuerua_to_farmgateua	NARDL	FarmGateUA	ProducerUA	218	0.003720540408719906	0.00451706675895454	-0.04257407029868335	0.002365958530547707	0.8578306153764598	0.9311147814862188	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.7207226886549168	0.0005881329649814298	7.718002014642544e-16	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::producer_prozor...	Вершки	Вершки	cream	nan	nan	nan	linear	initial	daily	pairwise_overlap	producer_prozorro	reverse	ProZorro	FarmGateUA	prozorro_to_farmgateua	NARDL	FarmGateUA	ProZorro	218	-0.001192557311013747	0.02020539430095059	-0.045339771192285044	0.001153436482021398	0.4706240117367897	0.6980113873401895	nan	unreliable	1	residual_autocorr_or_arch	0	0	0.8611020688741277	0.0001200750226678769	2.98769558098478e-15	Wald	short/long	asymmetry...	1	0	0	0
pairwise_product	pairwise::producer_prozor...	Молоко питне	Молоко питне	milk	nan	nan																																				